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Introductions

Instructor

- Name
- AspenTech Experience
- Educational Background
- Industry Experience
- Other

Student

- Name, company, and position
- Process simulation experience
- Expectations for the course

Course Objectives

At the end of this course, you will be able to:

- Develop the skills and techniques required for creating, modifying and running dynamic simulations
- Build dynamic models of vessel relief scenarios, compressor surge control and pipeline hydraulics
- Develop distillation column control with detailed geometry using the Column Analysis tool
- Apply the best practices for transitioning from steady-state to dynamic mode and discover shortcuts for efficient use of Aspen HYSYS Dynamics

Webinar Agenda

- Introduction to Aspen Hysys Process Simulator
- Introduction to Aspen Hysys Dynamics
- Transitioning from Steady State to Dynamic
- Compressor Surge Analysis

Where do I learn more? – AspenTech Training Offerings



Aspen Classroom Training

In person expert led classes, in traditional or virtual classrooms

Extensive On-Demand Training

Self-paced training of the same classroom material with intermittent instructor assistance



Aspen eLearning

- Includes videos, knowledge check quizzes, and exercises
- Accessible 24/7/365 from any location

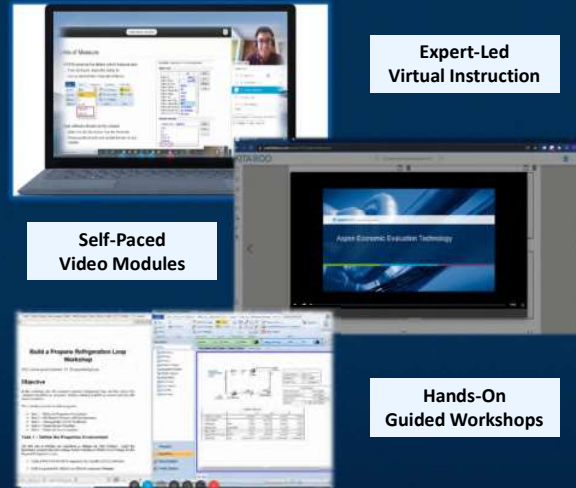


Aspen Certification

- First level certification available for users new to products
- Available for Aspen Plus, HYSYS, ACCE, PIMS, and DMC3. IP.21 and APM in development.

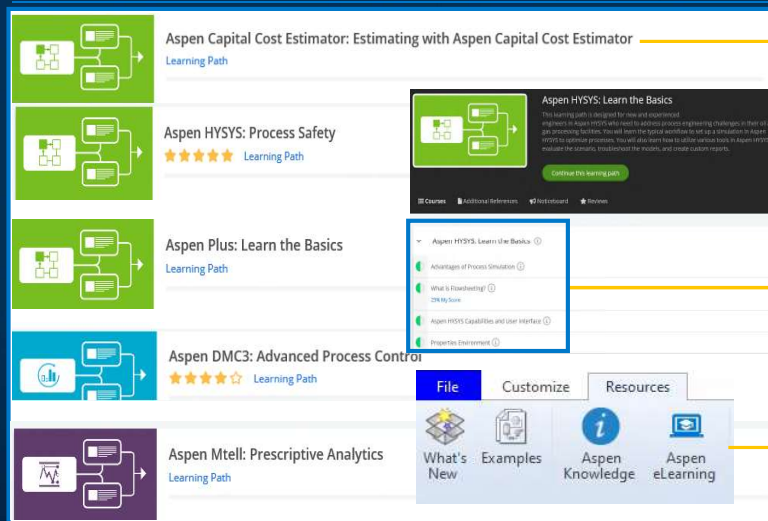
Extensive On-Demand Training

- Self-paced, rigorous training materials
- You control the speed of learning
- Learn from anywhere
- Hours worth of video lectures
- No sacrifice of training quality, no shrinking in training materials
- Instructor is available to you for the duration of the course



Aspen eLearning

LEARN. APPLY. SUCCEED



- Collection of learning paths (courses)
- Courses for ENG (green), MSC (blue) and APM (purple)
- Content equivalent to approximately 1 day of instructor led course
- Access 24/7/365 days from anywhere
- Each course contains modules with short 3-5-minute videos
- Quizzes at the end of most videos
- Practice with exercises
- Access eLearning from inside the Engineering products

AspenTech Certification Program



||| LEARN. APPLY. SUCCEED

What is it?

- Focused on users' proficiency on using the AspenTech solutions. Not designed to certify IT or engineering expertise
- More info on the [AspenTech Certification Website](#)
- Certification consists of the following key elements:



Proficiency Exams

Exam consisting 2 parts: multiple choice and hands-on exercises and have 4 hours to complete



Administering

Users take an exam in person or remote over the Internet administered by AspenTech



Language

All exams will be delivered in English unless otherwise requested by user



Credentials

Users who pass the exam and become certified will receive a digital certificate and badge to post on LinkedIn, valid for 3 years

How to Get Certified?



Step 1

Explore Certification

Choose a certification to pursue



Step 2

Prepare for Your Exam

Prepare for your certification by taking recommended training courses

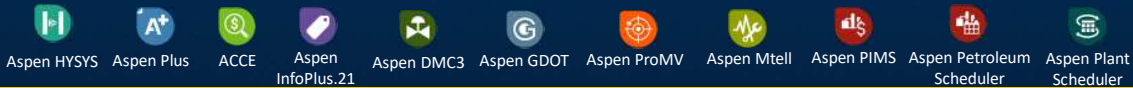


Step 3

Register for Your Exam

Once you identify which certification to pursue and are prepared, it's time to register

Certifications Available



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Pursue Different User Certification Levels



||| LEARN. APPLY. SUCCEED

Available



First-level credential that validates fundamental product skills and knowledge. ACU certifications address a wide spectrum of fundamental concepts, assess and validate core product knowledge, and enhance product credibility. Estimated Hours: 50 hours (includes training, using the product, and exam)

Available for some Products



Advanced-level credential that validates expert level product skills and knowledge. ACEU certification addresses advanced concepts, assess and validates advanced product knowledge and enhance product credibility. Estimated Hours: TBD hours (includes training, using the product, and exam)

TBD



Master level credential that validates the deepest level of product expertise. ACMU certification addresses mastery level concepts, assess and validates mastery level product knowledge and enhance product credibility. Estimated Hours: TBD hours (includes training, using the product, and exam)

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Complete the Course Evaluation — Thank you!

- You will receive an email with a link to the course evaluation, or you can fill out a hard copy provided by your Instructor.

The screenshot shows the AspenTech Support Center interface for a course evaluation. The course is 'EAP250 Aspen Plus: Distillation Modeling', taught by 'Instructor: [Name]', located in 'Virtual Americas', and running from 'August 3, 2020 - August 4, 2020'. The evaluation table has five columns: 'Very satisfied', 'Satisfied', 'Neither satisfied nor dissatisfied', 'Dissatisfied', and 'Very Dissatisfied'. The rows are: 'Sequence of course content and topics covered', 'Level of detail for topics presented', 'Effective mix of lectures and workshops', and 'Level of difficulty of workshop problems'. Each row has five radio buttons corresponding to the columns.

	Very satisfied	Satisfied	Neither satisfied nor dissatisfied	Dissatisfied	Very Dissatisfied
Sequence of course content and topics covered	☐	☐	☐	☐	☐
Level of detail for topics presented	☐	☐	☐	☐	☐
Effective mix of lectures and workshops	☐	☐	☐	☐	☐
Level of difficulty of workshop problems	☐	☐	☐	☐	☐

Complete the evaluation to download your certificate!



AspenTech Contact Information

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Training aspentech.training@aspentech.com
- Phone – Regional and in-country toll-free numbers at:
https://esupport.aspentech.com/Contact_Phone
- Support Site Tour:
https://esupport.aspentech.com/apex/S_Resources#sitetour
- Available training modalities:
<https://www.youtube.com/watch?v=t7EpEGvZWt4&t=27s>

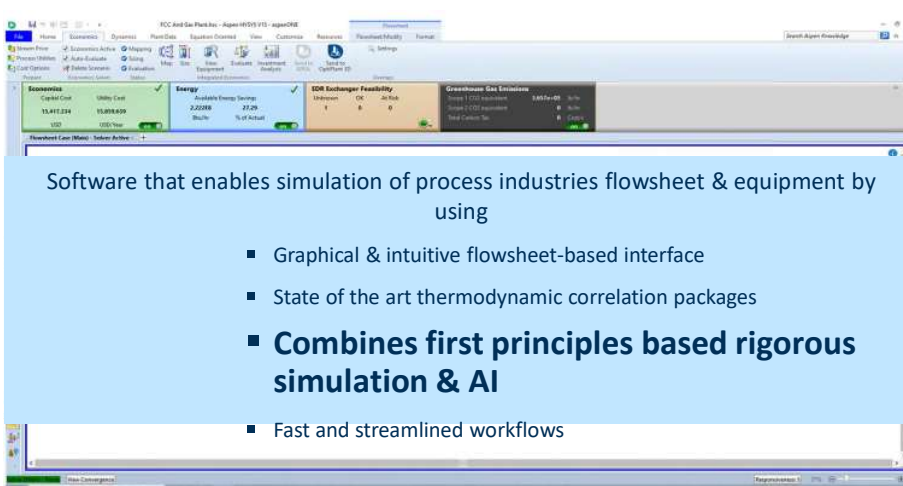


NALA

EMEA

APAC

Aspen HYSYS: The World's Leading Process Simulation Software



Software that enables simulation of process industries flowsheet & equipment by using

- Graphical & intuitive flowsheet-based interface
- State of the art thermodynamic correlation packages
- Combines first principles based rigorous simulation & AI**
- Fast and streamlined workflows

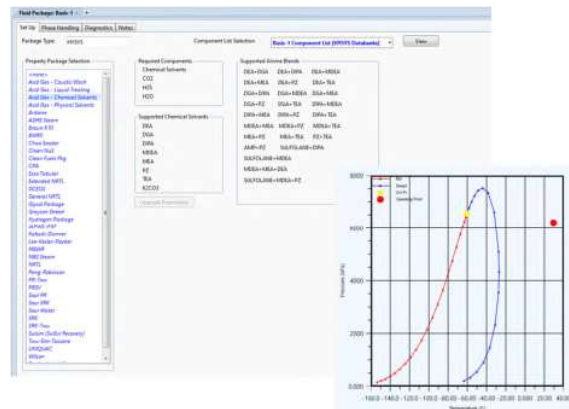


Leading Physical Properties Database Aspen HYSYS®

Model Processes with Confidence Using a Solid Foundation of First Principles

Capabilities

- Accurately predict process stream yields and properties by leveraging
 - Specialized packages for acid gas & carbon capture, dehydration, sour water treatment, sulfur recovery, H₂ liquefaction, etc.
 - The most complete set of physical properties data, built in collaboration with NIST
 - Wide range of methods and physical properties, including the predictive Peng-Robinson

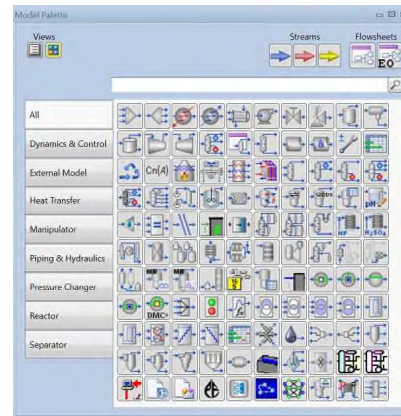


Intuitive Modeling of Unit Operations, Streams & Sub-flowsheets Aspen HYSYS®

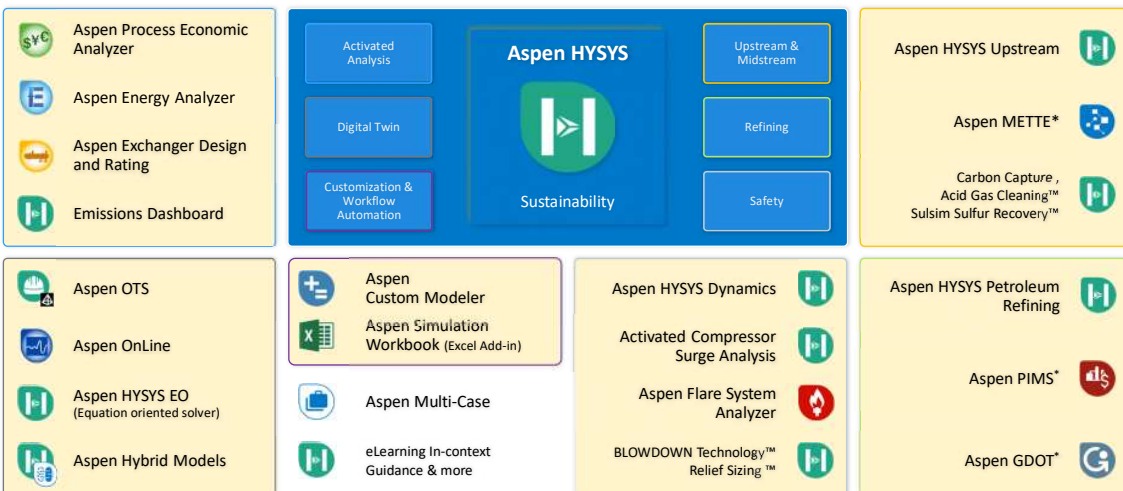
Easily Simulate Plant-Wide Processes with Off-the-Shelf Model Templates

Capabilities

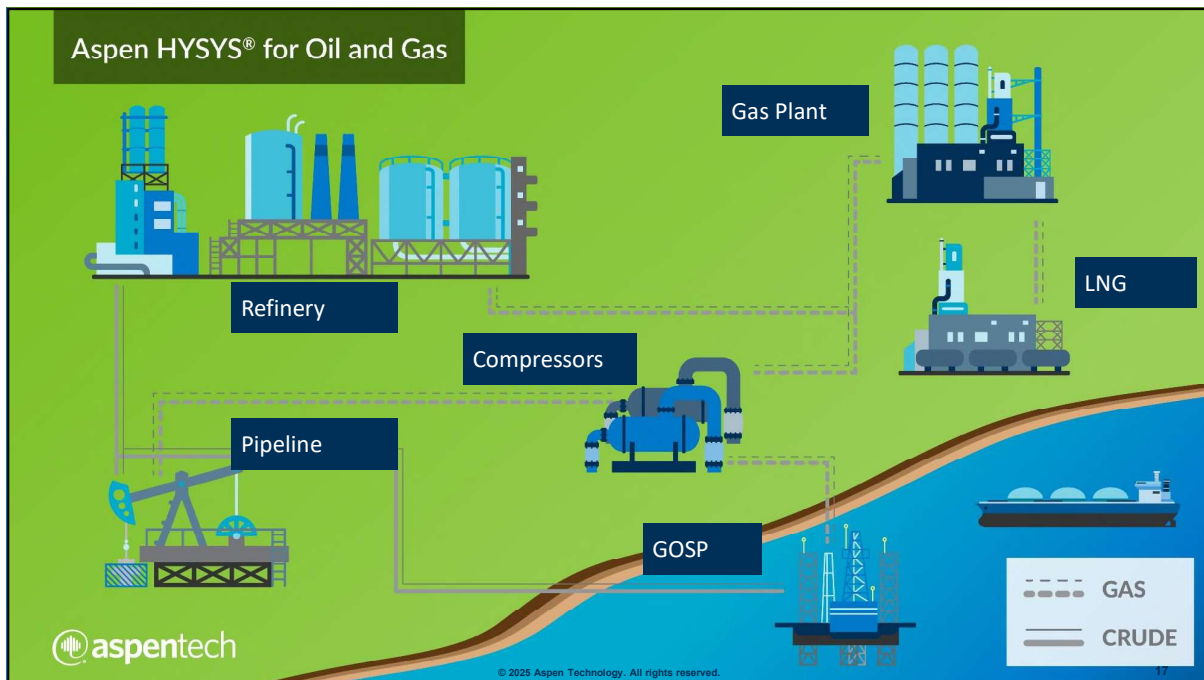
- Easily and intuitively build process flowsheet using pre-built templates for unit operations, streams and sub-flowsheets from model palette
- Model palette covers various
 - Streams
 - Reactors
 - Separator Equipment
 - Heat Exchangers
 - Pumps/compressors/ Control valves & more



Aspen HYSYS Engineering Family



* Not part of AspenTech's Performance Engineering Solution suite, but available with AspenTech's other solution suites like MSC, SSE etc.



Dynamic Studies

Aspen HYSYS®

Ensure Safety, Operability & Controllability

Benefits

- Minimize rework in detailed engineering
- Mitigate potential failures
- Improve operability & controllability

Dynamic Insights with Intuitive Setup

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Dynamic Modeling for APC

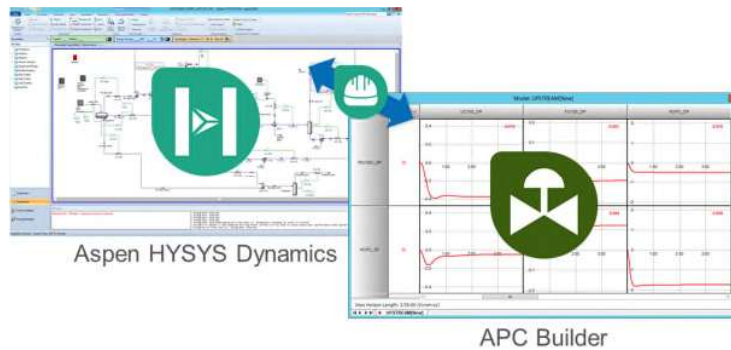
Aspen HYSYS Dynamics™

Reduce OPEX While Improving Yields & Quality

Benefits

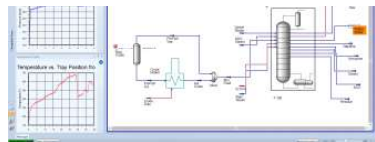
- Faster deployment of controllers by developing seed models for APC
- Test & improve control strategies using dynamic simulation
- Evaluate numerous scenarios without disturbances & other uncontrolled process behavior
- Enables dynamic simulation of processes with niche equipment simulated using hybrid models

Lifecycle Dynamic Modeling



Steady State Vs Dynamic Simulation

Steady State Simulation



Design/Rating/Optimization

Accumulation = 0

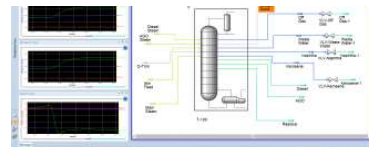
Linear Equations

Flow Driven

Modular non- sequential algorithm

No Process Control

Dynamic Simulation



Rating/Design Support/Validation

Accumulation $\neq 0$

Differential Equations

Pressure Driven

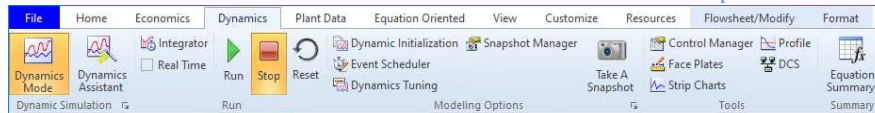
Modular sequential Algorithm

Process Control

Aspen HYSYS Dynamics™

- Dynamic process simulator integrated into Aspen HYSYS
 - Conservation equations account for process changes over time (material, energy, and composition)
 - Accumulation term differentiated with respect to time

$$\text{Accumulation} = \text{In} - \text{Out} + \text{Generation} - \text{Consumption}$$

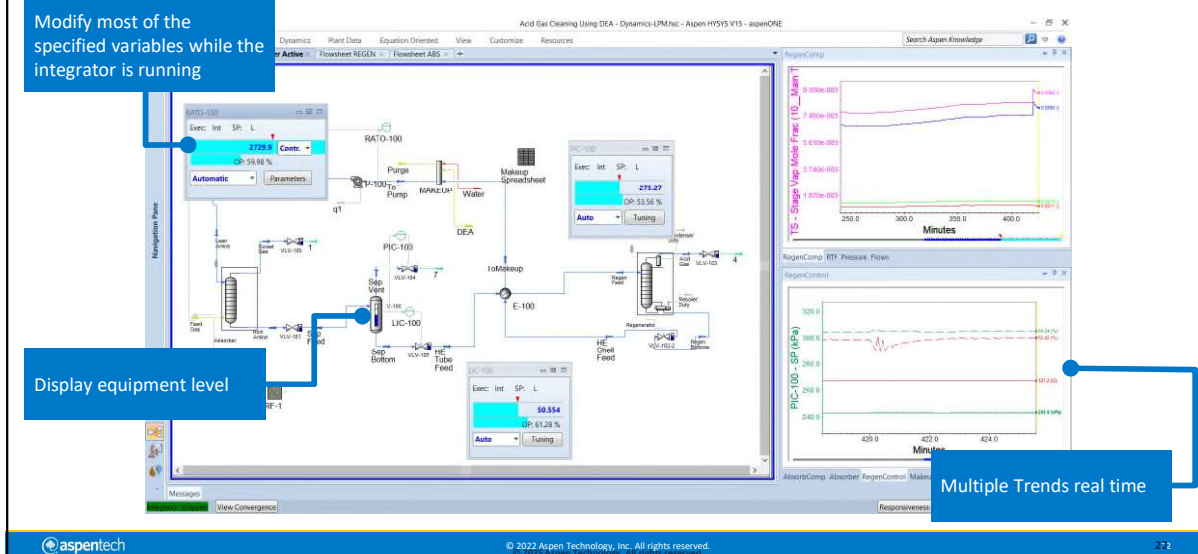


- Offers unique functionality not available through steady state operation alone
 - Event scheduler
 - Simulate startup, shutdown, disturbances, etc.
 - Process Control (PID, Ratio, MPC, etc.)
 - Pre-tune control loops, DCS checkout, etc.
 - Strip Charts
 - Data Trends, Exporting to Excel, etc.
 - Graphical UI tailored to analyze transient process
 - Equipment level indicators, differentiated valve types etc.

Aspen HYSYS Dynamics™ Interactive Environment

Modify most of the specified variables while the integrator is running

Display equipment level



Equipment Supported in Aspen HYSYS Dynamics

Rotating equipment

- Pumps (Positive displacement, centrifugal & polytropic)
- Compressors (Centrifugal and Positive displacement)
- Expanders

Static Equipment

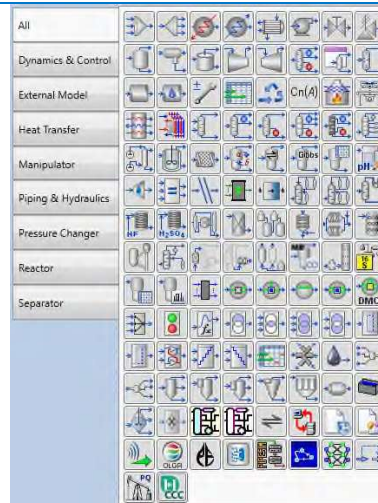
- Heat Exchangers (Shell & Tube, Air Cooler, Fired Heater)
- Tanks
- Reactors (CSTR, Plug flow, Conversion, Equilibrium, Kinetic)
- Columns

Valves

- Control Valves
 - Valve stations
- Pressure Relief Valves

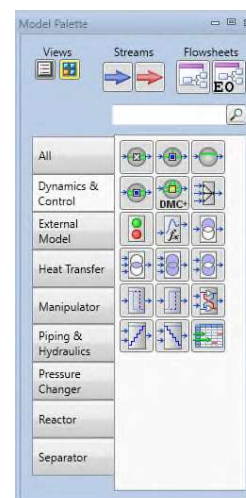
Pipe Segment

- Complex pipelines including pressure drop and heat transfer



Controller & Logical Operations Supported in Aspen HYSYS Dynamics

- Used to replicate control logic and strategies used in the actual plant DCS system
- Accessible in Dynamics & Control group of the Model Palette
 - Split Range Controller
 - Ratio Controller
 - PID Controller
 - Feedforward Controller
 - MPC Controller
 - Transfer Functions (Integrator, Lag, Lead, Delay, 2nd Order, Ramp, Rate Limiter)
 - Boolean (Not, And, Or, Xor, Off Delay, On Delay, Latch, Count Up, Count Down)
 - Cause & Effect Matrix



Key Applications of Dynamic Simulation

Safety System Evaluation

Operator Training Simulator

Operational Decision Support : Compressor Surge Analysis

A person is shown from behind, standing with their arms raised towards a large, glowing digital globe. The globe is composed of a network of white lines and dots, with various icons (like a factory, a lightbulb, and a gear) floating around it. The background is a bright, hazy sky with a sun or moon low on the horizon, creating a warm, golden glow. The overall image conveys a sense of reaching for technology or innovation.

Transitioning from Steady-State to Dynamic Mode

Section Objectives

At the end of this section, you will be able to:

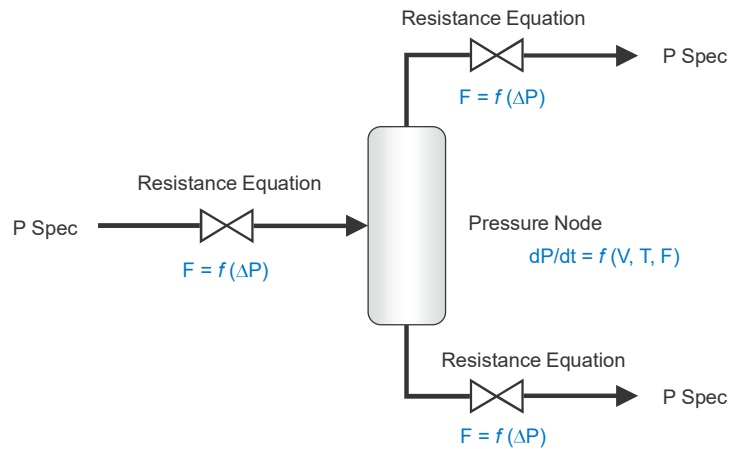
- Define dynamic pressure-flow specifications and enter equipment sizing data
- Understand the solving strategy and degrees of freedom analysis in Aspen HYSYS Dynamics
- Use the Dynamics Assistant tool to verify the preparedness of a simulation to be run in dynamic mode

Workshop: Convert a Steady-state Model into a Dynamic Model

Dynamic Solver

- The dynamic solver in Aspen HYSYS is not the same as the steady-state solver
- Aspen HYSYS Dynamics considers the flowsheet as a boundary value problem
- If pressure or flow (P/F) specs are made on all boundary streams (i.e., feed(s) & product(s)) then the P/F Solver will be able to solve for all internal pressures and flows simultaneously
 - Dynamic calculations are not sequential as in Aspen HYSYS steady-state
- Three Key Concepts:
 - Pressure-Flow specs need to be made on all boundary streams
 - Resistance equations / Conductance operations → Calculate a pressure drop and isolate boundary streams
 - Pressure node operations → Calculate pressure in process equipment

Pressure-Flow Solver



Dynamic Solver Execution

- Pressure-Flow equations that represent the flowsheet:
 - ▢ Resistance equations (dP-Flow)
 - ▢ Pressure nodes (P, V, T)
 - ▢ Material balances (with an accumulation term, differential equation)
- Control and logical operations
- Energy balance
- Streams composition and flash calculations
- The frequency of the calculations can be tuned (via the Integrator)

Every Time Step



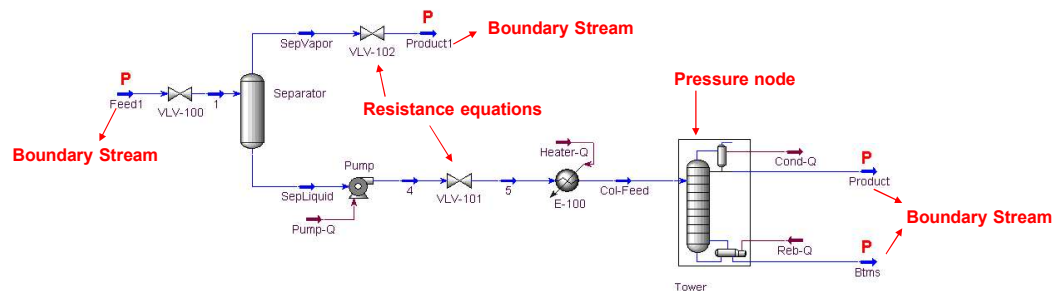
Less Often
(Configurable)

Dynamic Solver Procedure

- Aspen HYSYS Dynamics Solver uses a lumped system, not a distributed system
- Distributed system: Thermal and component gradients in 3D (x, y, z) are calculated with time
 - Requires partial differential equations in the 'x', 'y' and 'z' directions and 't' domain
 - Typically, these problems are solved via CFD
- Lumped system: x, y, z gradients are ignored, properties are considered equal in space
 - Represented by a set of ordinary differential equations (ODEs)
- The lumped method saves time and is reasonably close to the distributed model results
- Aspen HYSYS can also consider static head contributions for pressure calculations (dp/dz gradient considered in vessels)

Dynamic Model Setup

- In Aspen HYSYS Dynamics, a process simulation requires additional specifications in order to run in dynamic mode:
 - Dynamic P/F Specs need to be made on all boundary streams
 - Pressure node equations in equipment where accumulation (holdup) is allowed (e.g. vessels)
 - Resistance / Conductance equations to account for pressure drop in the process

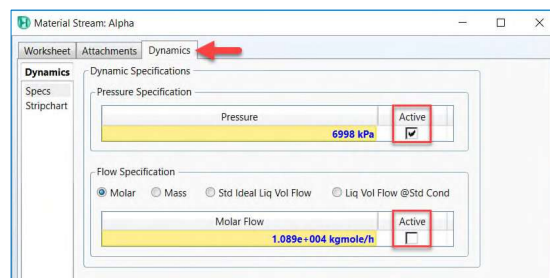


Equipment Sizing and Pressure Gradient

- Equipment sizing is very important in dynamic modeling - a dynamic simulation is a rating simulation
 - Review all unit operations to ensure they are ready to be run in dynamic mode
- Remember to pay attention to pressure drop across the plant
 - NO PRESSURE DROP = NO FLOW
 - POSITIVE PRESSURE GRADIENT = NEGATIVE FLOW (REVERSE FLOW)
- Consider your dynamic model as a real operating plant

Dynamic Pressure-Flow Specifications

- Start by making a dynamic pressure-flow specification on each boundary stream
 - Boundary streams are defined as flowsheet inlets or outlets
- Dynamic P/F Spec values are based on current steady-state results
 - Specs can be modified if active
- Only activate either pressure or flow, but not both!



Pressure Node Operations

- All unit operations with significant holdup volumes are pressure nodes in a dynamic simulation, examples:
 - Separators and tanks
 - Column tray sections
- Pressure node operations allow for the accumulation of material in the equipment
- The Pressure-Flow relationship can be defined as:

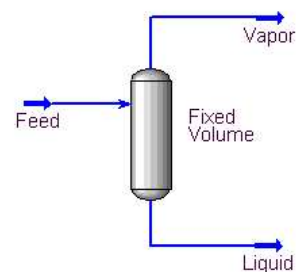
$$\frac{dP}{dt} = f(V, F, T)$$

Where: V = Fixed Volume, F = Flow, and T = Temperature

Pressure Nodes (1)

- A vessel has a fixed geometry (volume): $dV/dt = 0$
- With fixed volume, a pressure node is then calculated as a function of:
 - Vessel Temperature
 - Vessel Holdup
- In Dynamic mode, the rate of change in pressure is related to the rate of change of temperature (enthalpy) and the rate of change of material holdup (level):

$$\frac{dP}{dt} = f(V, F, T)$$

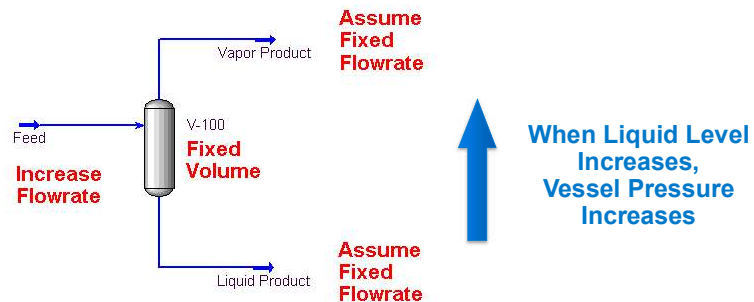


Pressure Nodes (2)

- The volumetric flow balance can be expressed as:

(Volume change due to pressure change) + (Volume change due to flow change) + (Volume change due to temp change) = 0

- For example:



Resistance / Conductance Equations

- The P-F relationship for these unit operations can be defined as:

$$Flow = k\sqrt{\rho\Delta P}$$

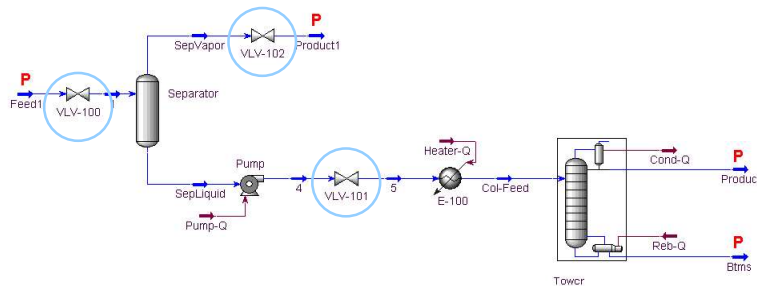
- The conductance ("k" term) is determined from steady-state conditions: flow, density and pressure drop
- Unit operations that use resistance equations include:
 - Heaters and coolers
 - Heat exchangers
 - PFRs
 - Pumps and compressors
 - Column tray sections

Resistance Equations - Valves

- For valves, Aspen HYSYS Dynamics uses the turbulent flow equation for the P-F relationship:

$$Flow = C_v \sqrt{\Delta P}$$

Note: The C_v proportionality constant involves the C_v of the valve and its opening %



Equipment Sizing Inputs

- Typically found on either the Rating or Dynamics tab
 - Valve sizing
 - Valve type
 - Normal valve opening position
 - Pressure drop across valve (from steady-state)
 - Current flowrate
 - Separator dimensions (volume, T-T length, diameter)
 - Heat Exchanger (volume, shell diameter, tube length)
 - Heater/Cooler (volume)
- Aspen HYSYS equipment sizing routines can be used, defaults accepted or values can be entered by users

Dynamic Inputs for Unit Operations

- You can use this table as a guide for transitioning unit operations to dynamic mode
- Available in the Help Menu Guide

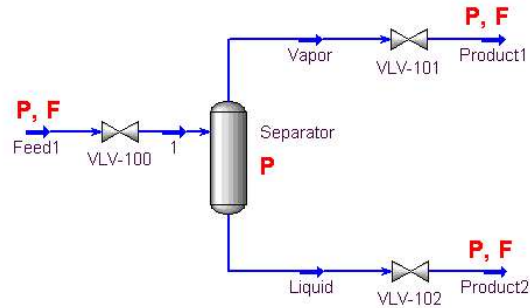
Dynamic Specifications	
Boundary Streams	Insert a valve on all boundary streams (feed/product streams) within the Flowsheet that are not connected to conductance devices (such as heat exchangers, coolers, heaters).
Pressure Specifications	Place a pressure specification on all boundary streams (feed/product streams) within the Flowsheet.
Distillation Column	Distillation columns with condensers require an extra specification around the condenser. Make a flow specification for the reflux flow.
Valves	Use the pressure/flow relationship as the dynamic specification for a valve.
K value	Use the overall K value as the dynamic specification for coolers, heaters, and heat exchangers and LNG exchangers.
Pressure Gradients	Be sure to account for pressure gradients throughout the Flowsheet and specify reasonable pressure drops/rises in the Flowsheet. Pressure differentials are the driving force for flow through the process Flowsheet.
Column Internals Analysis (New V9.0)	Use the Column Internals Analysis to estimate the column geometry and pressure profile.
Mixers	Use the Equalize All option as the pressure specification for mixers.
Tees	Remove Use Splits as Dynamic Flow Specs on tees.
Rotating Equipment (Pumps, Compressors, Expanders)	Use Efficiency and either Head or Pressure Rise as dynamic specifications for rotating Equipment, Compressor and Pump Curves , if available, make excellent dynamic specifications.
Hold-ups	Be sure to properly size equipment with hold-ups.
Dynamic specifications can only be modified when the Integrator is stopped. Once the Integrator is started the value of the dynamic specification can be changed (its value displays in blue), but the choice of dynamic specification cannot be changed.	

Solving Strategy

- Aspen HYSYS Dynamics uses an equation-oriented solver
 - Unlike Aspen HYSYS in steady-state, which uses a sequential modular solver
- A degrees of freedom (DOF) analysis is performed to determine if the proper number of inputs are in place - the system must be a square one
- Although the degrees of freedom analysis is important, it may not catch inconsistent specifications
 - A dynamic spec may satisfy the DOF analysis, but it could lead to an inconsistent solution (e.g., singularity problems)

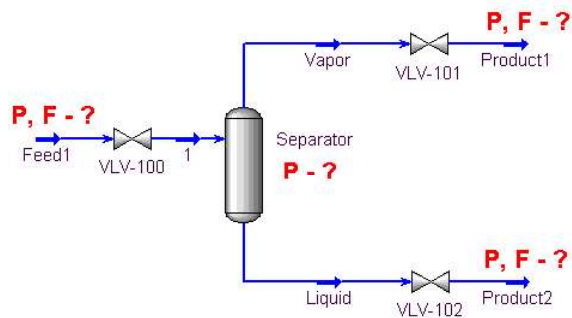
Degrees of Freedom Analysis

- 7 variables will define the system:
 - P, F of Feed1, Product1, and Product2 (6)
 - Separator Pressure (1)
- 4 equations define the flowsheet:
 - Turbulent Flow Equation:
 - $F_{valve} = Cv\sqrt{P_{In} - P_{Out}}$ (3)
 - Separator
 - $\frac{dP}{dt} = f(V, F, T)$ (1)
- DOF: $7 - 4 = 3$
 - The specification of 3 variables will completely define this system



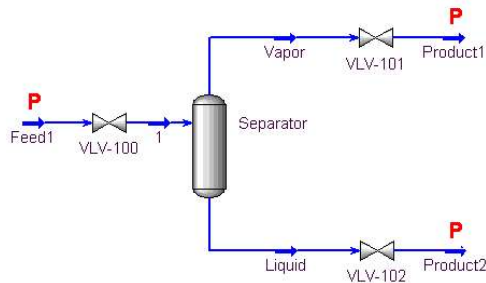
Available Dynamic Specifications

- A variety of specifications can be made:
 - Pressure specifications on material streams
 - Flow specifications on material streams
 - Fixed pressure drop across process equipment
 - Pressure-Flow equation (for valves)
 - Conductance calculations (for process equipment)



Acceptable Solution in Most Cases

- Pressure specifications made on all boundary streams
- Consistent solution
 - Vessel pressure calculated by holdup equation
 - Stream pressures calculated by resistance equations

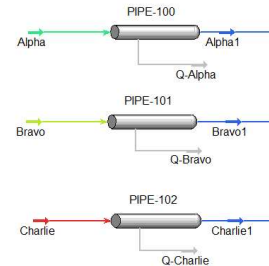


Guidelines / Rules of Thumb

- One P/F specification should be made on each boundary stream (i.e., inlets and outlets)
- Make pressure specifications on boundary streams attached to process equipment that use resistance to flow / conductance relationships
 - Analogy: feeds come from large storage tanks and products go to large storage tanks as well; tanks fix both the feed and product pressures
- Aspen HYSYS will use either the equipment conductance (K) or Cv valve value combined with the pressure of the inlet and outlet streams to determine the flowrate going through process equipment units
- Internal flowrates will be calculated by pressure gradients (resistance to flow / conductance equations) throughout the flowsheet

Dynamic P/F Specs Color Scheme

- You can select various color schemes for flowsheet streams
 - Dynamic P/F Specs scheme sets color based on pressure/flow specs
 - Other schemes are available
 - You can create your own scheme
- The Dynamic P/F Specs color scheme is extremely useful to visually check which streams have dynamic specs in place

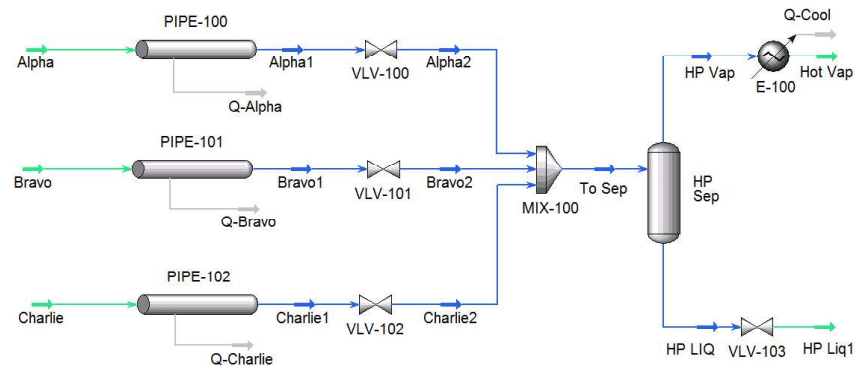


Dynamic Transition Summary

- Make pressure specs on all boundary streams
 - Generally do NOT make pressure nor flow specs on internal streams
 - Pressure specs are generally preferred over flow specs
- Add valves to boundary streams attached to separators
- Always size process equipment
- Enable P/F equations on valves / conductance operations
- Supply Cv and K values
- SAVE THE STEADY-STATE CASE BEFORE SWITCHING TO DYNAMIC MODE!

Workshop: Convert a Steady-State Model into a Dynamic Model

- Start with case 01-SteadyStateStarter.hsc, you will enter the necessary sizing information and make flowsheet specs in steady-state to jump straight to dynamic mode



Workshop: Convert a Steady-State Model into a Dynamic Model

This workshop includes the following tasks:

- Task 1 – Equipment Sizing
- Task 2 – Make Pressure/Flow Specifications
- Task 3 – Move to Dynamic Mode



Controllers and Strip Charts



Section Objectives

At the end of this section, you will be able to:

- Add and set up controllers in a dynamic model
- Create strip charts to monitor key process variables

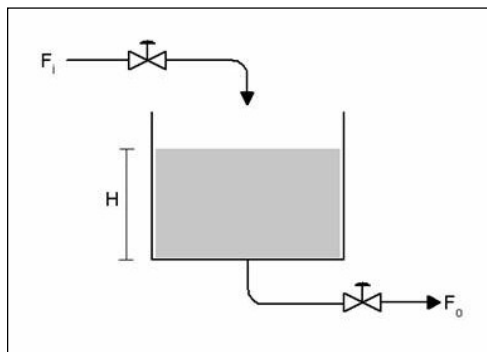
Workshop: Add Controllers and Strip Charts

Dynamic Modelling and Process Control

- Control of process variables such as flowrate, temperature, and pressure in a continuously operating plant attempts to:
 - Maximize profitability
 - Ensure product quality
 - Improve plant safety and operability
- While steady-state modelling allows for optimization of operating conditions, dynamic modelling allows to:
 - Design and test a wide variety of control strategies before implementation
 - Stress the system with disturbances and observe the response
 - Off-line comparison of alternative control strategies and operating schemes instead of disturbing the real process

Process Dynamics

- General conservation principle:
- Rate of Accumulation = Input – Output + Internal Generation – Internal Consumption

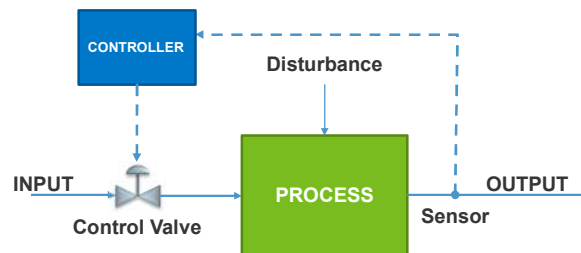


$$\frac{d}{dt} V = A \frac{d}{dt} H = F_i - F_o$$

System can be characterized
as first order

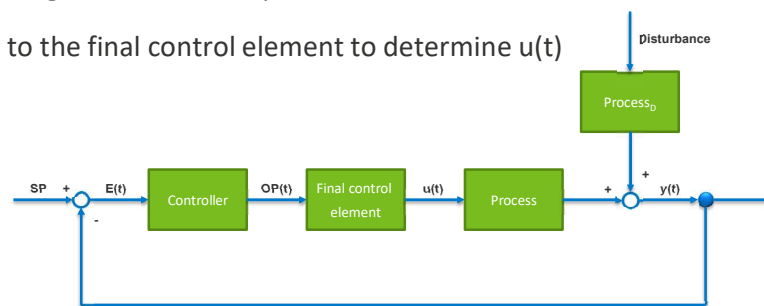
Closed-loop Control

- Aspen HYSYS Dynamics makes extensive use of closed-loop control systems
- Classic control equation:
 - K_c – Controller Gain
 - T_i – Reset Action (Min/Repeat)
 - T_d – Derivative Action



Feedback Control Block Diagram

- Measure the controlled variable, PV
- Compare the PV value against the setpoint (SP); calculate error ($E(t)$)
- Supply $E(t)$ to the general control equation; OP% is calculated
- OP% is passed to the final control element to determine $u(t)$

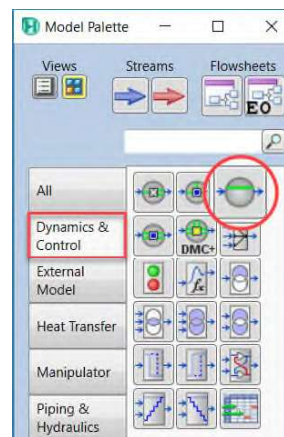


Factors That Affect Closed-loop Stability

- Controller tuning parameters
- Non-linearities in the process
- Non-linearities in the instrumentation
- Interactions between control loops
- Frequency of disturbances
- Capacity of process (ability to absorb disturbances)
- Noise in the measurement of process variables

PID Controllers

- PID Controllers can be added to the flowsheet the same way as for other unit operations
- Once the controller has been added:
 - Select the process variable (PV) source
 - Set both the minimum and maximum bounds for the PV
 - Select the output (OP) target object
 - Specify the controller action: reverse or direct
 - Enter the controller tuning parameters (k_c , t_i , t_d)
 - Choose the controller mode:
 - Off, manual, automatic, indicator



PID Controller Configuration

The image displays two screenshots of the AspenTech FC-Alpha PID Controller configuration interface. The left screenshot shows the 'Connections' tab, which includes a process flow diagram with a circle representing the controller. A red box highlights the 'Face Plate...' button, and a red arrow points to it. The right screenshot shows the 'Parameters' tab, which is divided into 'Operational Parameters' and 'Tuning Parameters'. In the 'Operational Parameters' section, the 'Action' is set to 'Reverse' (indicated by a red arrow). Other parameters include 'SP Mode' (Remote), 'Mode' (Casc), 'Execution' (Internal), 'SP' (2.676e+005 kg/h), 'PV' (2.676e+005 kg/h), and 'OP' (50.52 %). The 'Tuning Parameters' section includes 'Kc' (0.250), 'Ti' (0.100), 'Td' (<empty>), 'Tt' (<empty>), 'b' (1.00), and 'c' (0.000). The 'Algorithm Type' is set to 'Hysys' and the 'Algorithm Subtype' is 'PID Velocity Form'.

Controller Action

- There are two types of controller action:
 - Reverse action:
 - If the PV increases above the SP, the OP of the controller will decrease to drive the PV down
 - Example: Steam control
 - Direct action:
 - If the PV increases above the SP, the OP of the controller will increase to drive the PV down
 - Example: Level control
- Choosing the right action is critical!

The image shows a screenshot of the AspenTech TIC-100 @CDL1 PID Controller configuration interface. The 'Parameters' tab is active, displaying 'Operational Parameters'. The 'Action' is set to 'Reverse' (indicated by a red arrow). Other parameters include 'SP Mode' (Local), 'Mode' (Auto), 'Execution' (Internal), 'SP' (68.01 C), 'PV' (65.44 C), and 'OP' (60.11 %). The 'Tuning Parameters' section includes 'Kc' (0.500), 'Ti' (3.00), 'Td' (<empty>), 'Tt' (<empty>), 'b' (1.00), and 'c' (0.000). The 'Algorithm Type' is set to 'HYSYS' and the 'Algorithm Subtype' is 'PID Velocity Form'.

Tuning Parameters – Rules of Thumb

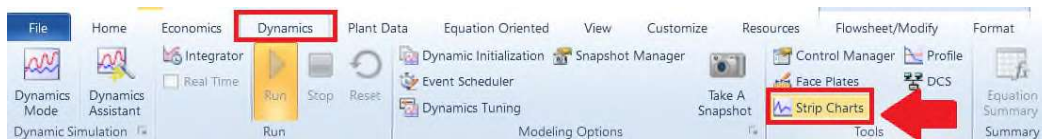
System	K_c	T_i (min)	T_d (min)
Flow	0.1	0.2	0
Level	2	10	0
Pressure	2	2	0
Temperature	1	20	0
Composition	0.1	0.2	0

Other Aspen HYSYS Dynamics Controllers

- **Ratio Controller** 
 - Special type of feedforward controller designed to keep a constant ratio between two variables: manipulated and load
- **Split Range Controller** 
 - Uses multiple manipulated variables, all driven by the same controller, to control a single process variable
- **Digital Point Controller** 
 - On/off, dead band-type controller; similar to a thermostat
- **MPC (Model Predictive Controller)** 
 - Used in processes that are inherently multi-variable and many process parameters interact and affect process control

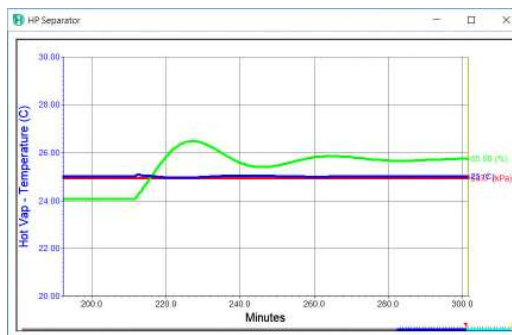
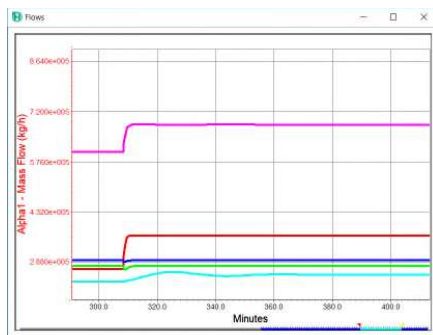
Strip Charts

- Strip charts (plots) are a feature that allows you to graphically monitor the response of process variables in the dynamic model
 - The behaviour of variables is captured graphically as calculations are dynamically performed
- Accessible from the Dynamics tab



Strip Chart Configuration

- Multiple Strip Charts can be built
 - For clarity and ease of reporting, a maximum of six variables per chart is recommended
 - Create multiple strip charts to monitor different sections of your dynamic process model



Strip Chart – Historical Data

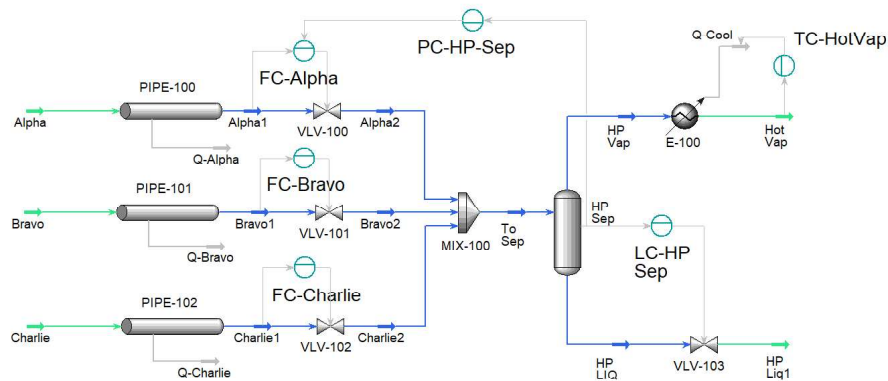
- Once a strip chart has been set up and the model run in dynamic mode, historical data related to the process variables included in a strip chart can be accessed
 - Historical data can be exported as either a *.csv or *.dmp file or you can even use the copy-paste Windows functionality to bring those data to another application (e.g., MS Word)

Time	Alpha1 - Mass Flow (kg/h)	Bravo1 - Phase - Mass Flow (Overall) (kg/h)	Charlie1 - Mass Flow (kg/h)	Hot Vap - Mass Flow (kg/h)
62316.0	2.669e+005	2.774e+005	2.932e+005	6.051e+005
62336.0	2.669e+005	2.774e+005	2.932e+005	6.051e+005
62356.0	2.669e+005	2.774e+005	2.932e+005	6.051e+005
62376.0	2.669e+005	2.774e+005	2.932e+005	6.051e+005
62396.0	2.669e+005	2.774e+005	2.932e+005	6.051e+005
62416.0	2.669e+005	2.774e+005	2.932e+005	6.051e+005
62436.0	2.669e+005	2.774e+005	2.932e+005	6.051e+005

☒ Ascending Order

Workshop: Add Controllers and Strip Charts

- Start with the final case from Workshop 1 and add controllers and strip charts to create a fully integrated dynamic process model



Workshop: Add Controllers and Strip Charts

This workshop includes the following tasks:

- Task 1 – Add PID Controllers
- Task 2 – Build Strip Charts and Observe the Dynamic Behavior of the Model
- Task 3 – Customize Strip Charts and Change the Controllers Set Point



Section Objectives

At the end of this section, you will be able to:

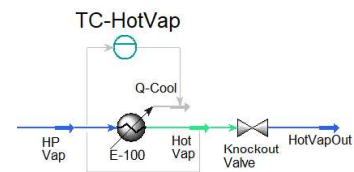
- Add unit operations directly in dynamic mode
- Initialize newly added unit operations in dynamic mode
- Develop appropriate control strategies using split range and on-off controllers
- Install a relief valve for vessel overpressure protection
- Explore Dynamics Tuning Manager capabilities

Adding Unit Operations directly in Dynamic Mode

- Dynamic results must be present
 - Used to initialize new unit operations and streams
- Add new unit operations to the flowsheet
- Size the unit operation
 - If previous results exist, they can be used to size the operation
 - If no results exist, users will need to enter sizing data
- Add boundary valves (if needed)
 - If a boundary stream is attached to a conductance operation (e.g., heater, cooler, PFR) a valve might not be necessary at all

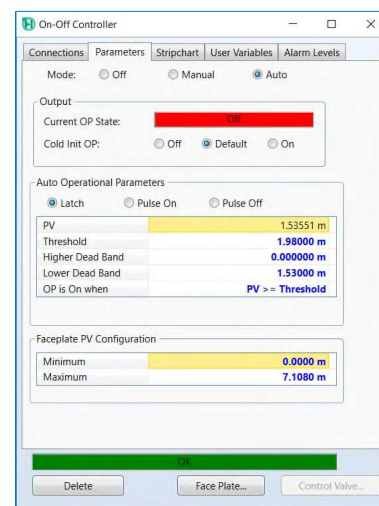
Building Models and Moving Dynamic Specs

- Add additional controllers (if needed)
- Move P/F specs to the “new” boundary stream if possible
 - A P/F spec on a boundary stream is not an absolute requirement but a good simulation practice
 - For example: keep a P/F spec on a valve inlet stream and assume a reasonable dP to size the valve
 - Run the model in dynamic mode for a few seconds to calculate the valve outlet pressure and the pause it
 - Swap the P/F spec from the valve inlet stream to the outlet stream
 - Run in dynamic mode for a few seconds to stabilize the model



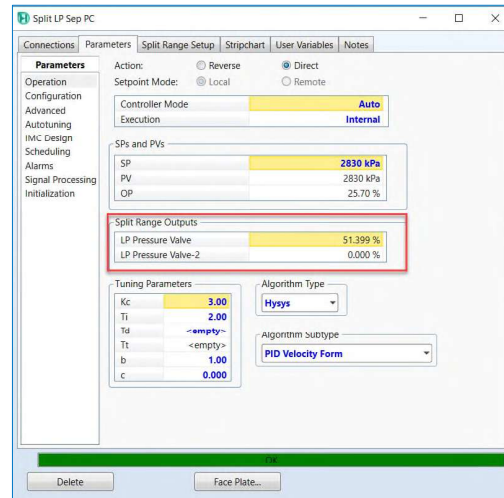
On-Off Controller

- Referred to as Digital Point operation
- Select a process variable (PV) to monitor and an object to control (OP) it with
- When the PV reaches the defined threshold, the digital point turns on / off the OP
 - The Dead Band input allows to specify a range of PV values to indicate when the OP should be on / off



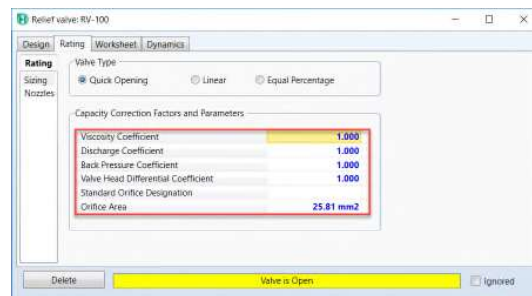
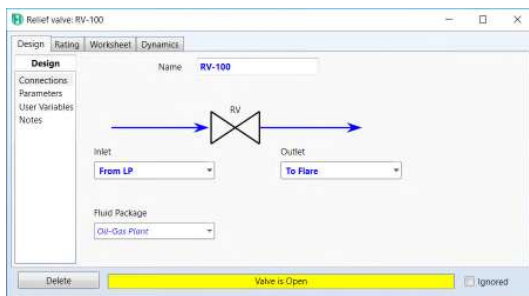
Split Range Controller

- Allows to set multiple manipulated variables (OP) to control a single process variable (PV)
- Examples:
 - Two valves for pressure control on a three-phase separator
 - Pressure control in a reactor by controlling both inlet and outlet flowrates



Relief Valve

- Used to model any type of relief valve in Aspen HYSYS Dynamics
- The relief valve will open once the pressure in the process equipment exceeds the defined set pressure



Dynamics Tuning Manager (1)

- The Dynamics Tuning Manager (implemented in V11) provides options for modifying your dynamic model, helping ensure uninterrupted integrator performance.



Dynamics Tuning Manager (2)

- One of the most common scenarios for pressure-flow solver failure is when a section of the process has zero flow.
- Define a section of the flowsheet and set up actions that will be triggered when a failure-prone scenario is encountered. For example, select a section of the process that will be ignored when it handles zero flow.
- Such section will be automatically restored when it gets back to normal or a more stable operation.



Compressor Curves and Surge Control



Section Objectives

At the end of this section, you will be able to:

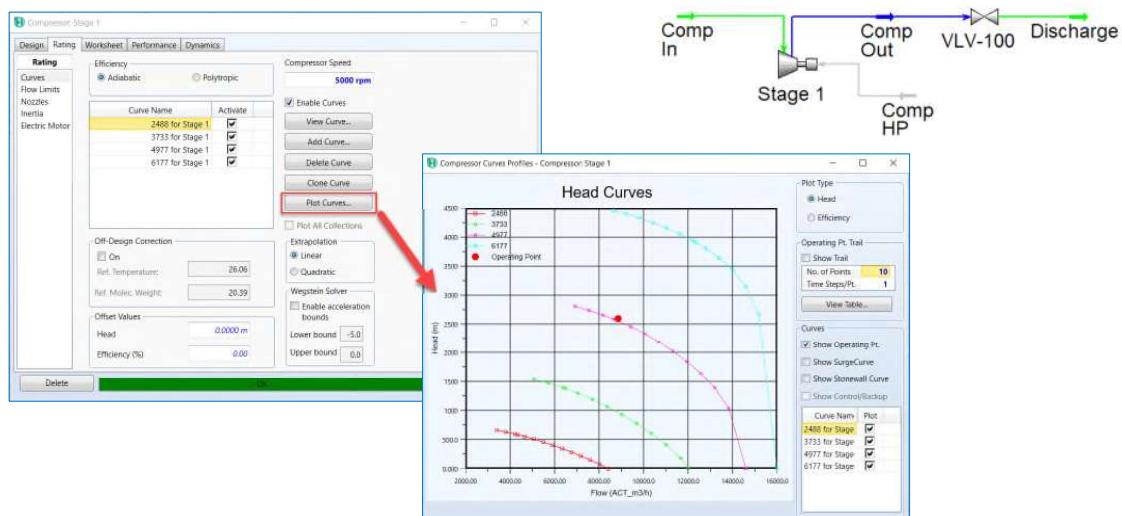
- Robustly model compressors using performance curves
- Model rotating equipment in dynamic mode
- Implement surge control schemes for compressors

Workshop: Compressor Curves and Surge Control

Compressor Surge Control

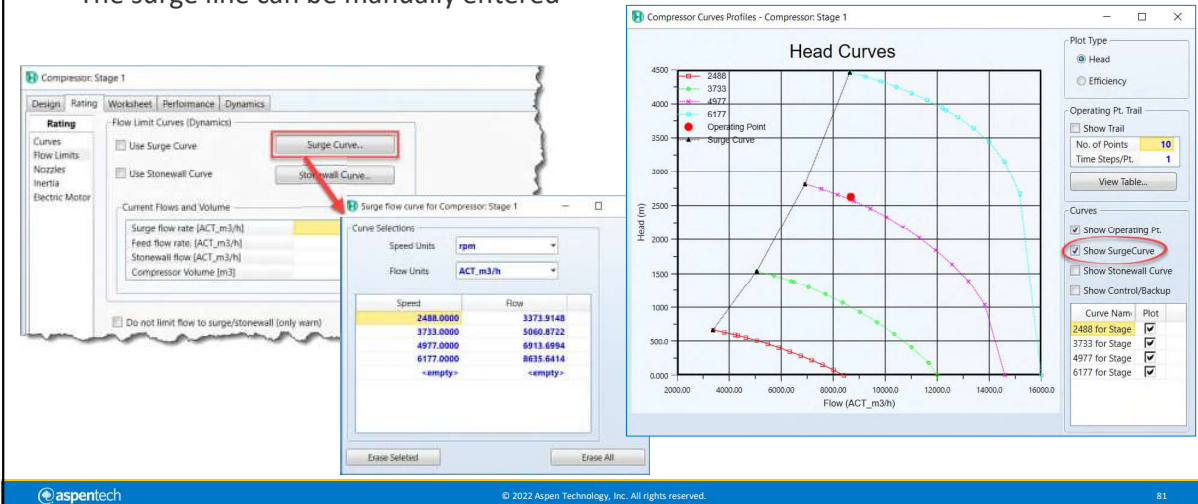
- If a centrifugal compressor goes below a certain throughput at a given head, the performance of the compressor becomes erratic and the vibrations associated with this erratic behaviour can be extremely harmful for the compressor
- To protect the compressor, a recycle valve is typically installed to make sure there will always be a minimum throughput
- The controller for this strategy must also react very quickly when surge conditions are about to be met, but remain smooth and robust when the compressor is operating normally

Compressor Curves



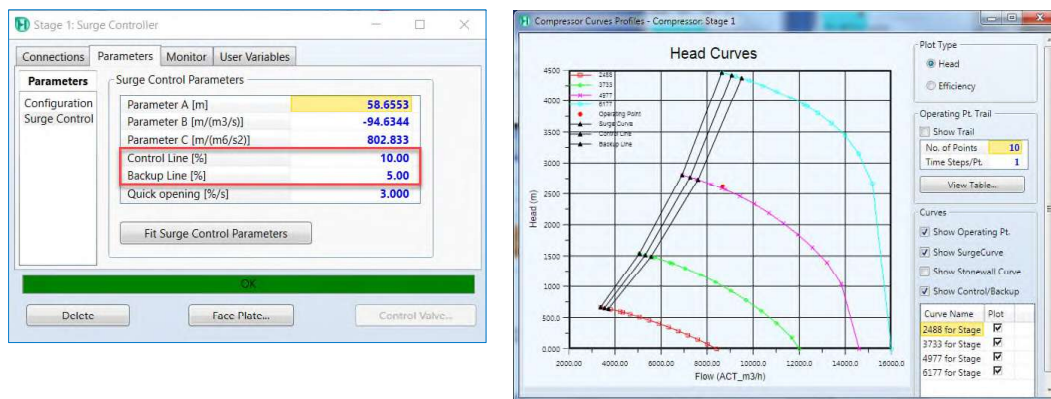
Entering the Surge Line Data

- The surge line can be manually entered



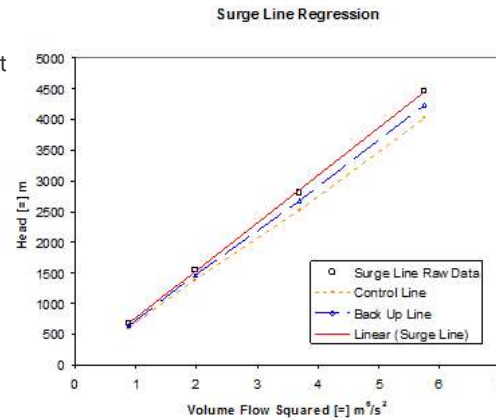
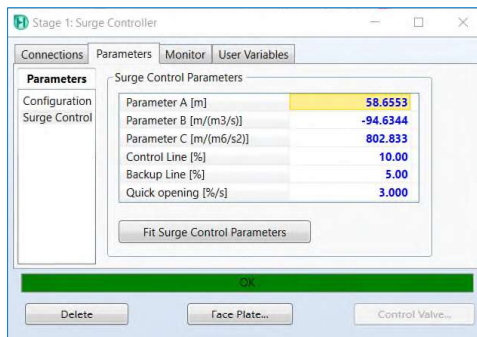
Control Line and Backup Line

- Both the control and backup lines can be displayed along with the surge line



Regressing the Surge Line

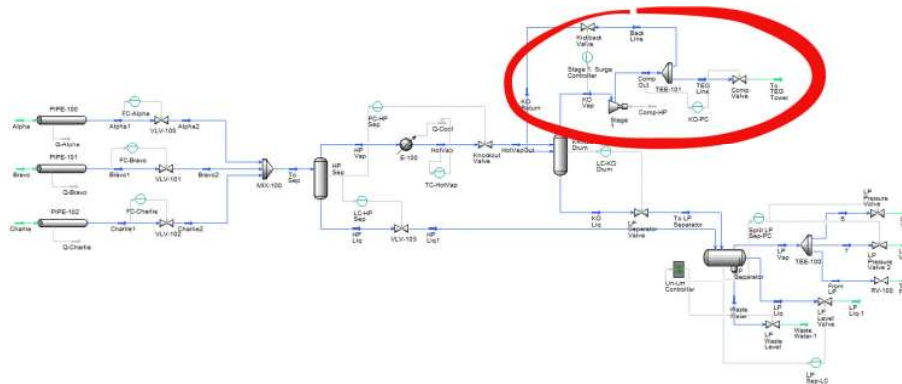
- User-regressed surge controller parameters
 - Regressed from low-flow performance curve point



$$\text{Head} = A + B \cdot \text{Flow} + C \cdot \text{Flow}^2$$

Workshop: Compressor Curves and Surge Control

- Add a compressor using performance curves, and set up an anti-surge control loop to protect the compressor against low-flow scenarios



Workshop: Compressor Curves and Surge Control

This workshop includes the following tasks:

- Task 1 – Import the Compressor Model
- Task 2 – Set up the Compressor
- Task 3 – Add the Surge Controller



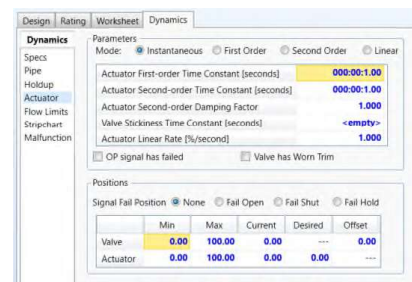
Dynamic Details

Valve Characteristics

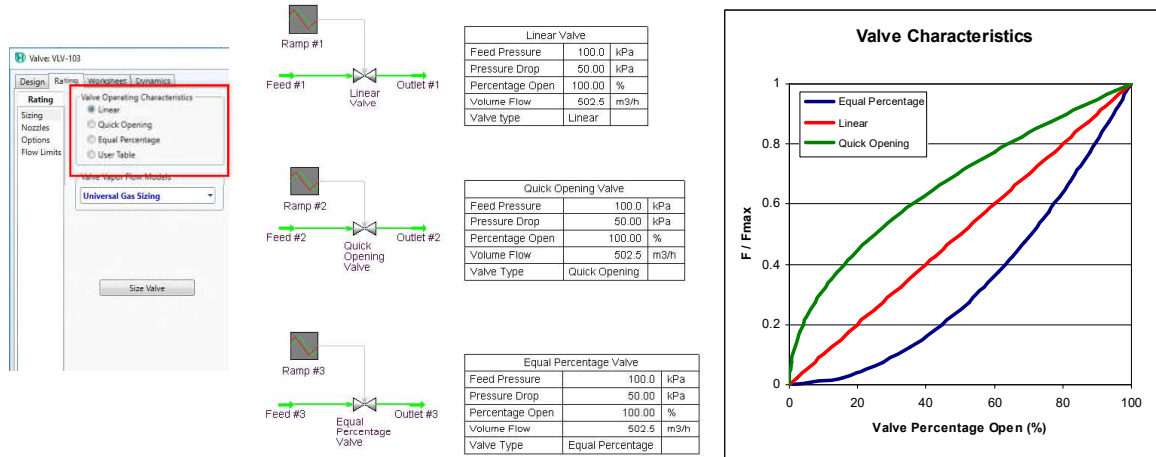
- Aspen HYSYS Dynamics allows users to rigorously model valves with several options:
- Actuator Rate:
 - Instantaneous
 - First Order (Actuator Time Constant)
 - Linear (Actuator Linear Rate)
- Stickiness Time Constant
- Leaky Valves (Min value position)
- Valve Operating Characteristics

Introduction to Control Valves

- One of the most important parts of a control loop
- Successful process control strategies which involve control valves require:
 - Selection procedure
 - Correct sizing
- By default, Aspen HYSYS valves operate ideally:
 - Instantaneous actuator
 - Fully opens and closes
 - Actuator does not stick
- Modify the valve actuator to depart from ideal behavior



Valve Operating Characteristics



Integrator Setup

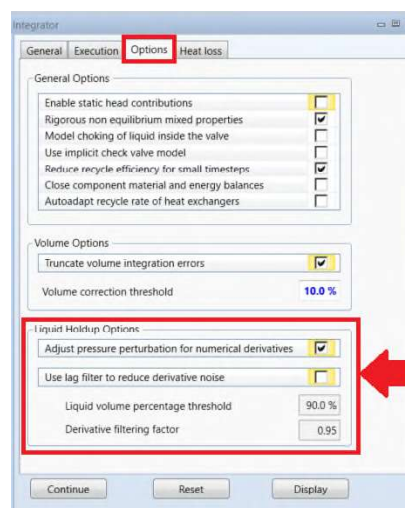
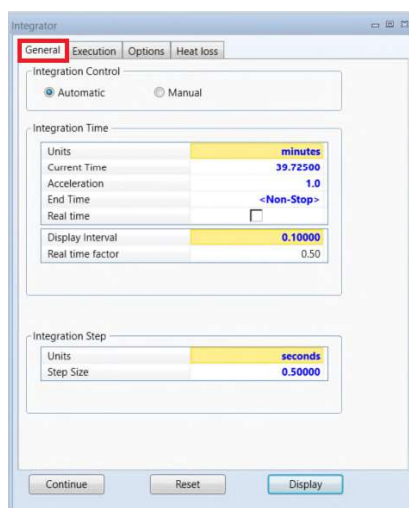
- The integrator runs the simulation when switching to dynamic mode
 - Accessible from the Dynamics tab in the ribbon
- It performs the integration of pressures / flows, enthalpies, and composition – it uses three different time step sizes:
 - Simultaneous P/F relationships → Smallest time step
 - Logical ops and enthalpy (energy) balances → Intermediate time step
 - Composition calculations → Largest time step
 - Option to modify the frequencies of each dynamic calculation
- Integration Method
 - Fully Implicit Euler integration method
 - Proved to be stable, robust and fast



Integrator Options

- When the Integrator is running, values for the P/F specs can be changed; however, the structure of specifications may not be changed
 - When the Integrator is stopped the structure of P/F specs as well as their values can be changed
- Static head contributions can be enabled in the Integrator form
 - Static head is a function of:
 - Equipment holdup
 - Nozzle elevation difference
- Hint: Ctrl+I opens the Integrator form

Integrator Parameters



New Feature in V11